# MODULE 4 – DBMS (Theory Answers)

## 1. Introduction to SQL

### *Q1. What is SQL, and why is it essential?*

SQL (Structured Query Language) is used to manage and manipulate relational databases. It helps in storing, retrieving, updating, and deleting data efficiently and ensures data integrity and security.

### *Q2. Difference between DBMS and RDBMS*

DBMS stores data in files, while RDBMS stores data in tables with relationships. RDBMS supports normalization, constraints, and better security.

### *Q3. Role of SQL*

SQL is used to create databases, insert data, update data, delete data, retrieve data, and manage user permissions.

### *Q4. Key features of SQL*

Easy syntax, data integrity, security, large data handling, and transaction support.

## 2. SQL Syntax

### *Q1. Basic components*

Keywords, tables, columns, clauses, and conditions.

### *Q2. SELECT structure*

SELECT column_name FROM table_name WHERE condition;

### *Q3. Role of clauses*

WHERE filters, ORDER BY sorts, GROUP BY groups data.

## 3. SQL Constraints

### *Q1. What are constraints?*

Rules to ensure data integrity. Types include PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE.

### *Q2. PRIMARY KEY vs FOREIGN KEY*

Primary key uniquely identifies rows; foreign key links tables.

### *Q3. NOT NULL & UNIQUE*

NOT NULL prevents empty values; UNIQUE prevents duplicates.

## 4. DDL

### *Q1. What is DDL?*

DDL defines database structure.

### *Q2. CREATE command*

Used to create database/table.

### *Q3. Importance*

Ensures proper data storage and integrity.

## 5. ALTER

### *Q1. Use*

Modify table structure.

### *Q2. Operations*

Add, modify, drop columns.

## 6. DROP

### *Q1. Function*

Deletes table/database permanently.

### *Q2. Implication*

Data cannot be recovered.

## 7. DML

### *Q1. INSERT, UPDATE, DELETE*

Used to manipulate data.

### *Q2. WHERE importance*

Prevents affecting all rows.

## 8. DQL

### *Q1. SELECT*

Retrieves data.

### *Q2. ORDER BY & WHERE*

WHERE filters, ORDER BY sorts.

## 9. DCL

### *Q1. GRANT & REVOKE*

Manage permissions.

### *Q2. Managing privileges*

Controls user access.

## 10. TCL

### *Q1. COMMIT & ROLLBACK*

COMMIT saves changes, ROLLBACK undoes changes.

### *Q2. Transactions*

Ensures safe execution.

## 11. JOINS

### *Q1. Types*

INNER, LEFT, RIGHT, FULL JOIN.

### *Q2. Use*

Combine multiple tables.

## 12. GROUP BY

### *Q1. GROUP BY*

Used with aggregate functions.

### *Q2. Difference*

GROUP BY groups, ORDER BY sorts.

## 13. STORED PROCEDURE

### *Q1. What*

Predefined SQL block.

### *Q2. Advantage*

Reusable and efficient.

## 14. VIEW

### *Q1. View*

Virtual table.

### *Q2. Advantage*

Simplifies queries.

## 15. TRIGGER

### *Q1. Trigger*

Auto-executed SQL.

### *Q2. Types*

INSERT, UPDATE, DELETE.

## 16. PL/SQL

### *Q1. PL/SQL*

Extension of SQL.

### *Q2. Benefits*

Supports logic and programming.

## 17. CONTROL STRUCTURES

### *Q1. IF & LOOP*

Used for logic.

## 18. CURSOR

### *Q1. Cursor*

Fetch row-by-row data.

## 19. SAVEPOINT

### *Q1. SAVEPOINT*

Used for partial rollback.